Phase 2 - Grade 2 - Calculus and Gradients

Key Learning Points -- Derivatives

A **derivative** answers: 'if I nudge the input a tiny bit, how much does the output change?' -- the instantaneous rate of change at a specific point. Car analogy: position changes over time; speed (derivative of position) is how fast position changes right now.

f'(x) / df/dx notation

The prime mark (') simply means 'the derivative of'. f'(x) and df/dx mean the same thing -- the rate of change of f with respect to x.

Power rule

For f(x) = x^n, f'(x) = n · x^(n-1) -- bring the exponent down as a multiplier, reduce the exponent by 1. n and x are read directly off the function as written -- n is whatever exponent is already there; x stays symbolic unless a specific value is later substituted in as a separate final step.

x^0 = 1 rule

Any nonzero number to the power 0 equals 1 -- needed when n=1 in the power rule (x^(1-1) = x^0 = 1).

Constant rule

If f(x)=c (a plain number, no x), f'(x)=0 -- a constant never changes as x changes.

Sum rule

If f(x) = g(x) + h(x), differentiate each term separately then add: f'(x) = g'(x) + h'(x).

Average rate of change

(f(b)-f(a)) / (b-a) over a GAP between two points. Works for any gap size, tiny or large; a large gap blends the whole interval into one number.

Instantaneous rate of change

The exact rate at ONE point, no gap -- what the average rate of change approaches as the gap shrinks toward zero. This IS the derivative.

Derivation: Average Rate of Change -> Derivative Formula

`f(x) = x^2. Compare x to a nearby point x+h.`

`f(x+h) = (x+h)^2 = x^2 + 2xh + h^2`

`[f(x+h)-f(x)] / h = [2xh+h^2]/h = 2x + h (average rate of change, general formula)`

`As h -> 0: 2x + h -> 2x => f'(x) = 2x (the derivative -- the formula, actually derived)`

Key Learning Points -- Partial Derivatives & Gradients

A real function often depends on MULTIPLE variables (e.g. price depends on size AND location). A **partial derivative** asks: nudge ONE variable, hold every other variable frozen/constant, how does the output change?

Do this once per variable, then bundle all the results into one vector -- that vector is the **gradient**.

Analogy that made it click: standing on a hillside, altitude = f(x,y) where x=east-west position, y=north-south position. Walking east changes altitude at one rate (∂f/∂x); walking north changes it at a DIFFERENT rate (∂f/∂y). The gradient bundles both rates together: it's not one single steepness number, because steepness genuinely differs by direction.

Partial derivative (∂f/∂x)

The derivative with respect to ONE variable, treating every other variable as a fixed constant. Mechanically: freeze the other variable(s) at 'some' fixed value, then differentiate normally -- any term containing ONLY the frozen variable becomes a constant and drops to 0 via the constant rule.

Gradient (∇f)

All of a function's partial derivatives bundled into one vector: ∇f = [∂f/∂x, ∂f/∂y, ...].

Worked Example: Partial Derivatives & Gradient

```
f(x,y) = x^2 + y^2

∂f/∂x (freeze y): x^2 -> 2x; y^2 (frozen) -> 0. ∂f/∂x = 2x

∂f/∂y (freeze x): x^2 (frozen) -> 0; y^2 -> 2y. ∂f/∂y = 2y

∇f = [2x, 2y]

Numeric check at x=3, y=4 -- nudging x only (y fixed at 4): rate ≈ 6, matches 2x=2(3)=6

Nudging y only (x fixed at 3): rate ≈ 8, matches 2y=2(4)=8
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Key Learning Points -- Chain Rule

Some functions are NESTED -- one function feeding into another, e.g. f(x) = (x^2+1)^3. The **chain rule** handles this: differentiate the outer function (using a placeholder for the inner part), differentiate the inner function alone, then MULTIPLY the two results together.

Analogy that made it click: salary depends on rating, rating depends on hours -- a chain, hours→rating→salary. The total effect of hours on salary = (effect of rating on salary) × (effect of hours on rating). Each link's rate multiplies through the chain -- rates compound sequentially like gears, not like adding bricks.

Chain rule formula

d/dx[f(g(x))] = f'(g(x)) · g'(x) -- derivative of the outer function (evaluated at the inner function) times the derivative of the inner function.

The 4-step placeholder method

1) Name the inner function u=g(x). 2) Differentiate the OUTER shape using u as a placeholder (f=u^n -> df/du=n·u^(n-1)), then substitute back what u is. 3) Differentiate the INNER function alone, du/dx, as a plain ordinary derivative -- completely separate from the outer function. 4) Multiply Step 2's result by Step 3's result.

Worked Example: Chain Rule

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f(x) = (x+4)^2
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Step 1: u = x+4 (inner function)
Step 2: f = u^2 -> df/du = 2u -> substitute back -> 2(x+4)
Step 3: du/dx for u=x+4 -> 1 + 0 = 1
Step 4: f'(x) = df/du × du/dx = 2(x+4) × 1 = 2(x+4)

Numeric grounding (salary example): rating=hours×2, salary=rating^2
Link 1 (d(rating)/d(hours)) = 2. Link 2 (d(salary)/d(rating)) = 2·rating = 12 at rating=6
d(salary)/d(hours) = Link2 × Link1 = 12 × 2 = 24 (verified by direct nudge calculation: 24.04 ≈ 24)
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Clarified Doubts

My doubt: What does the ' (prime) symbol mean, and how are x and n determined in a power-rule problem?

Clarification: ' means 'the derivative of' -- f'(x) and df/dx are the same thing. n is the exponent already written in the function; x stays symbolic (a placeholder) unless the problem separately asks to evaluate at a specific number, which is always a second, later step -- never done at the same time as finding the general formula.

My doubt: How does the two-point (x, x+h) numeric calculation actually relate to the derivative formula?

Clarification: The average rate of change over a gap h, computed symbolically as [f(x+h)-f(x)]/h, simplifies to an expression like 2x+h. As h shrinks toward 0, that expression approaches 2x -- which IS the derivative formula. The formula isn't a separate fact; it's the exact numeric process, generalized with symbols, with the gap shrunk all the way to zero.

My doubt: Does 'average rate of change' only apply to tiny gaps, or can it be computed over a large interval too?

Clarification: It works for ANY gap size. A tiny gap gives a number very close to the instantaneous rate at that one point. A large gap (e.g. x=2 to x=5) gives a valid but coarser number that blends the entire interval together -- it does not reveal the instantaneous rate at any single point within that range.

My doubt: What is the actual use of the constant rule, if it's always just zero?

Clarification: Constants show up ATTACHED to real functions (e.g. an intercept in y=mx+b). The constant rule is what lets you correctly say that attached constant contributes nothing to the rate of change and can be dropped, rather than wrongly assuming it must do something in the derivative.

My doubt: In the hillside example f(x,y), what do x and y actually represent, and can you explain partial derivatives with real numbers instead of just abstractly?

Clarification: x and y represent position (east-west, north-south); f(x,y) is altitude at that position. Using real numbers (position x=3,y=4, nudging each direction separately by 0.01) showed that walking east changes altitude at a different rate than walking north -- proving one single number can't capture the hill's steepness; you need one rate per direction, which is exactly what a partial derivative computes.

My doubt: What does 'two separate questions, two separate derivatives, bundled together into the gradient' mean?

Clarification: The two questions are 'how fast if I only nudge x?' and 'how fast if I only nudge y?'. Each is answered by its own partial derivative (∂f/∂x and ∂f/∂y). 'Bundled into the gradient' just means packing those two separate numbers into one vector, ∇f = [∂f/∂x, ∂f/∂y] -- the same idea as a vector bundling multiple numbers into one 'thing'.

My doubt: In the chain rule, what exactly does Step 2 ('differentiate the outer function, leaving the inner part as one block') mean?

Clarification: Give the inner function a temporary placeholder name, e.g. u = x^2+1. The outer function becomes simply f=u^3 -- now it's an ordinary power-rule derivative using u instead of x: df/du=3u^2. Only AFTER taking this derivative do you substitute the real inner formula back in place of u.

My doubt: What exactly does Step 3 ('differentiate the inner function alone') mean?

Clarification: Forget the outer function entirely. The inner function (e.g. u=x^2+1) is just an ordinary function on its own -- take its plain derivative using rules already known (power rule, sum rule, constant rule), completely independent of whatever outer function it later feeds into.

Examples

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# Derivatives
f(x) = x^3           -> f'(x) = 3x^2;         f'(3) = 27
f(x) = x              -> f'(x) = 1x^0 = 1
f(x) = x^2 + x        -> f'(x) = 2x + 1
f(x) = 4x^3 + 6       -> f'(x) = 12x^2
f(x) = 2x^2 + 3x + 10 -> f'(x) = 4x + 3;      f'(2) = 4(2)+3 = 11

# Partial derivatives & gradient
f(x,y) = x^2 + y^2    -> df/dx = 2x, df/dy = 2y,  grad f = [2x, 2y]

# Chain rule
f(x) = (x^2+1)^3
  u = x^2+1;  df/du = 3u^2 = 3(x^2+1)^2;  du/dx = 2x
  f'(x) = 3(x^2+1)^2 * 2x = 6x(x^2+1)^2
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Quiz Q&A;

Q1. Find f'(x) for f(x)=5x^3+2x+8, then evaluate f'(2).

A: f'(x) = 15x^2 + 2. f'(2) = 15(4)+2 = 60+2 = 62.

Q2. Given f(x,y)=4x^2+3y^2, find ∂f/∂x, ∂f/∂y, and the full gradient ∇f.

A: ∂f/∂x = 8x. ∂f/∂y = 6y. ∇f = [8x, 6y].

Q3. Given f(x)=(x+5)^2, use the chain rule to find f'(x).

A: u=x+5. df/du=2u=2(x+5). du/dx=1. f'(x) = 2(x+5) × 1 = 2(x+5).

Q4. What is the difference between an average rate of change and an instantaneous rate of change? When does one approximate the other?

A: Average rate of change measures change over an interval/gap between two points. Instantaneous rate of change measures the exact rate at one single point. The average approximates the instantaneous rate when the interval is made very small (shrinking toward zero), or exactly equals it when the rate of change is

perfectly constant.

Q5. Why does a partial derivative treat other variables as constants? Tie it to the hillside/salary example.

A: Treating other variables as constants isolates the impact of a single variable on a multivariable system -- e.g. on the hillside, it answers 'if I hike strictly due east from where I'm standing right now, how steep is my immediate path?', ignoring the north-south direction entirely for that question.

Q6. Why does the chain rule require multiplying the outer and inner derivatives, rather than adding them?

A: Rates of change compound sequentially through a chain, like gears or multipliers, rather than stacking independently like bricks -- each link's rate scales the next link's effect (e.g. hours' effect on rating, then rating's effect on salary, cascades multiplicatively through the chain).